

EdTech Student Activity Project

STAGE 1 - BUSINESS UNDERSTANDING

Objective:

The objective of this project is to analyse student engagement and learning behaviour on an EdTech platform. The analysis aims to identify patterns and factors that influence student engagement, course completion, dropout risk, and overall learning outcomes.

STAGE 2 – DATA CLEANING & PREPROCESSING

2.1 Initial Data Quality Check

The dataset was inspected to identify missing values, duplicate student records, inconsistent categorical values, invalid dates, and incorrect numerical data types before analysis.

Data Quality Check	Action Taken
Missing Values	Identified and handled using appropriate imputation methods
Duplicate Student IDs	Checked and cleaned
Categorical Values	Standardized spaces, capitalization, and special characters
Numerical Values	Missing values filled using median
Date Values	Converted to datetime and missing dates forward-filled
Engagement Score	Validated within the expected range
Session Time	Checked for invalid negative values
Completion Flag	Missing values filled using mode

2.2 Handling Missing Numerical Values

Missing values in numerical student activity and engagement fields were filled using the median of the respective column. Median imputation was selected because it is less affected by extreme observations. The selected numerical columns were then converted to integer format.

Numerical Column	Treatment
login_frequency	Median + Integer
time_spent_hours	Median + Integer
videos_watched	Median + Integer
quizzes_attempted	Median + Integer
assignments_submitted	Median + Integer
forum_posts	Median + Integer
engagement_score	Median + Integer
dropout_risk_score	Median + Integer
avg_session_time	Median + Integer
inactive_days	Median + Integer
satisfaction_rating	Median + Integer

2.3 Categorical Data Standardization

Categorical columns were cleaned by removing leading and trailing spaces, removing unwanted special characters, standardizing capitalization, and filling missing values with the mode of each respective column.

Column	Cleaning Method
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device_type	Standardized and missing values filled with mode
location	Standardized and missing values filled with mode
subscription_type	Standardized and missing values filled with mode
payment_status	Standardized and missing values filled with mode
referral_source	Standardized and missing values filled with mode

For example, values such as “mobile”, “Mobile”, and “ MOBILE ” were standardized to “Mobile”.

2.4 Handling Duplicate Student Records

The student_id column was checked for duplicate values. Duplicate records were cleaned to prevent repeated students from affecting calculations, aggregations, and visualizations.

Metric	Final Result
Duplicate Student IDs	0

2.5 Student ID Validation

The student_id column was treated as a unique identifier. Forward filling was not used because it could assign the same ID to multiple students. A missing student ID was instead assigned a new unique ID using the next available identifier.

Validation	Result
Missing Student IDs	0
Duplicate Student IDs	0

2.6 Course ID Missing Value Handling

Missing course IDs were handled using mode imputation as required for the project. The most frequently occurring course ID was used to replace missing values, and the column was converted to integer format.

Metric	Result
Missing Course IDs Before Cleaning	1,296
Missing Course IDs After Cleaning	0
Imputation Method	Mode

2.7 Completion Flag Cleaning

The completion_flag column was checked for missing values and incorrect data type. Missing values were replaced using the mode and the column was converted from float to integer. A separate descriptive status can represent the values as Completed and Not Completed.

Value	Meaning
1	Completed
0	Not Completed

2.8 Date Conversion and Forward Filling

The last_login_date column was converted to datetime format. Missing or invalid dates were handled using forward fill (ffill), where each missing date was replaced with the most recent valid date above it. This retained the records without generating random dates.

Validation	Result
Date Column	last_login_date
Data Type	Datetime
Missing Dates After Forward Fill	0

2.9 Numerical Data Type Standardization

After missing-value treatment, the selected numerical columns were converted to integer format to maintain consistent data types for analysis, SQL processing, and Power BI aggregation.

Check	Final Treatment
Missing Numerical Values	Median imputation
Numerical Data Type	Integer

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2.10 Final Data Quality Validation

A final validation was performed after cleaning. Missing values, duplicate student IDs, data types, date values, categorical consistency, engagement score range, and session time values were checked before proceeding to analysis.

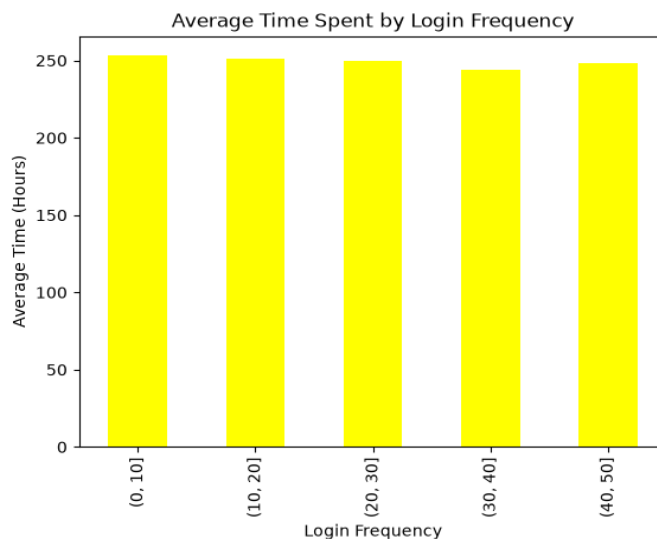
2.11 Final Dataset After Cleaning

Metric	Final Result
Rows	18,601
Columns	20
Duplicate Student IDs	0
Remaining Missing Values	0
Invalid/Missing Login Dates	0
Numerical Columns	Standardized
Categorical Columns	Standardized

STAGE 3 – EXPLORATORY DATA ANALYSIS

Objective: To analyse the cleaned EdTech student activity data(18,601 students) and understand how learners engage with the platform across login frequency,time spent,device type and location.The analysis also examines course completion,satisfaction ratings and inactive days.The goal is toturn these findings into clear,data backed actions that improve student retention ad course completion.

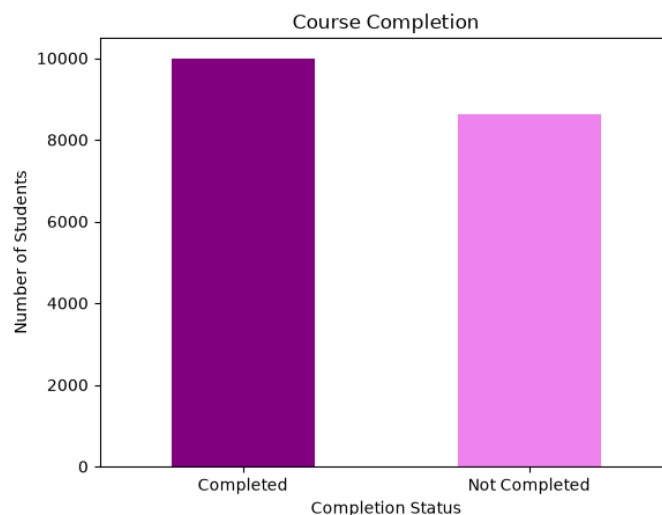
3.1 Average Time Spent by Login Frequency



Students are grouped into login-frequency bins of 10 (0-10 up to 40-50) and the mean of `time_spent_hours` is plotted for each bin.

- The average time spent is roughly 245-255 hours in every login bin.
- Logging in more does not mean studying more, Students who log in 40-50 times spend about the same total hours as those who log in 0-10 times.
- Login count alone is a weak proxy for effort. Session length (`avg_session_time`) and content activity (videos, quizzes) are likely better engagement signals

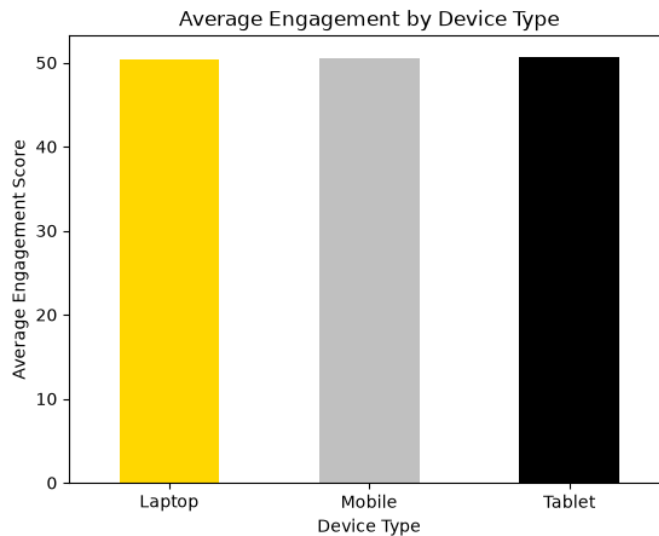
3.2 Course Completion



Vertical bar chart of category counts (Completed vs Not Completed).

- More completers than non-completers, Roughly 10,000 students (about 54%) completed versus roughly 8,600 (about 46%) who did not.
- nearly half of the learners do not finish, which is a significant retention gap for an edtech platform.
- missing `completion_flag` values were filled with the mode (Completed), which may slightly overstate the completion rate.

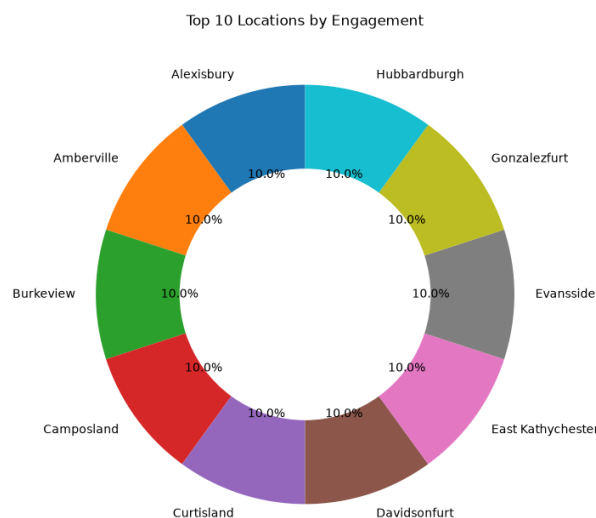
3.3 Average Engagement by Device Type



comparing the mean engagement_score across three categories (Laptop, Mobile, Tablet).

- All three devices sit at about 50-51 on the engagement score.
- Device type does not explain engagement, so the platform appears to deliver a consistent experience on laptop, mobile and tablet.

Average Engagement by Locations

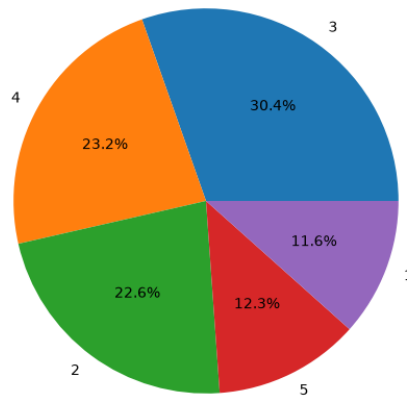


Of the ten locations with the highest mean engagement_score.

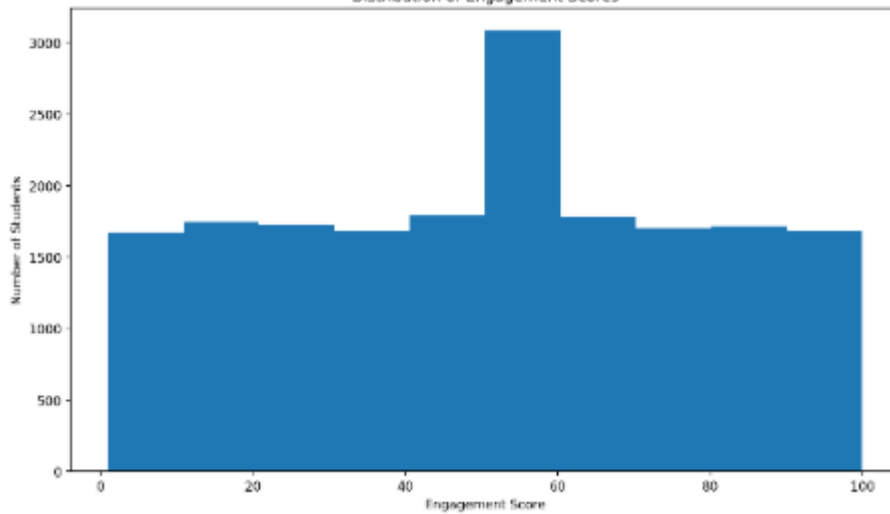
- All ten slices are exactly 10%, The ten locations have near-identical average scores, so the chart cannot rank them.
- The dataset has about 12,500 distinct locations for 18,600 students, so most locations contain only one or two students. A location's average is then just one person's score, and the top ten are simply students at the top of the scale.

3.4 Correlation between Engagement score and Satisfaction

Distribution of Satisfaction Ratings

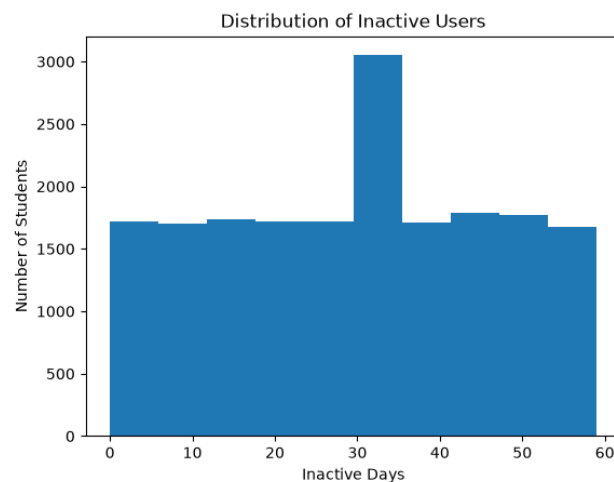


Distribution of Engagement Scores



- Rating 3 is the largest share (30.4%), followed by 4 (23.2%) and 2 (22.6%). Ratings 5 (12.3%) and 1 (11.6%) are the smallest.
- Satisfied students (4-5) make up about 35.5% and dissatisfied students (1-2) about 34.2%, so satisfaction is split evenly rather than clearly positive.

3.5 Distribution of Inactive Days



many students fall into each range of inactive_days (0 to about 59 days).

- About 1,700 students sit in each bin, so inactivity is evenly from 0 to 59 days.
- Students inactive for 30+ days form about half of the base and are a natural target for re-engagement campaigns.

EDA VISUALS THAT HAVE BEEN USED

S. No.	Chart / Analysis	Chart Type
1	Time Spent by Login Frequency	Vertical Bar Chart (Binned)
2	Course Completion	Vertical Bar Chart
3	Engagement by Device Type	Vertical Bar Chart
	Locations by Engagement	Donut Chart
4	Satisfaction Ratings	Pie Chart
5	Inactive Days	Histogram (10 Bins)

STAGE 4 - EdTech SQL Analysis

1: Top Engaged Students based on Activity

Output: The 10 highest engagement scores, each with the student ID. The top score is 100, and 78 students share it, so these 10 are a sample of the tied group.

student_id	engagement_score
296	100
402	100
551	100
555	100
711	100
876	100
879	100
908	100
961	100
1353	100

2: Dropout Rate by Subscription Type

Output: Dropout rate is about 46% under both plans. Premium is slightly higher than Free (46.51% vs 46.09%), so subscription type makes almost no difference.

subscription_type	dropout_rate (%)
Premium	46.51
Free	46.09

3: Average Session Time by Device

Output: Average session time is about 62 on every device, with Tablet slightly the highest (62.64) and Laptop the lowest (61.95).

device_type	avg_session_time
Laptop	61.95
Mobile	62.17
Tablet	62.64

4: Completion Rate by Location

Output: The query returns 11,892 locations, so the first 10 are shown. Most locations have only one or two students, so their rates are mostly 0%, 50% or 100%

location	completion_rate (%)
North Matthewbury	0.00
North Reneeburgh	100.00
West Robertville	66.67
North Kennethmouth	0.00
East Valeriestad	0.00
Port Allen	100.00
Lake Brian	71.43
Jasonmouth	0.00
Reginachester	100.00
Jessicahaven	100.00

5: High-Risk but Low-Activity Students

Output: The query returns 1,464 students (about 7.9% of all students) who are high risk and low activity, so the first 10 are shown. These are the students most in need of a retention intervention.

student_id	dropout_risk_score	engagement_score
1	100	9
19	97	22
31	98	3
33	92	23
39	70	19
42	94	15
43	90	20
50	90	5
79	96	10
83	79	5

6: Monthly Active Users Trend

Output: Active students per month from 2025-09 to 2026-09. Activity is steady at about 1,450-1,650 students per month from 2025-10 to 2026-08, and the first and last months are lower because they are partial months.

month	active_users
2025-09	385
2025-10	1,521
2025-11	1,583
2025-12	1,606
2026-01	1,516
2026-02	1,459
2026-03	1,538
2026-04	1,478
2026-05	1,589
2026-06	1,527
2026-07	1,631
2026-08	1,630
2026-09	1,138

7: Engagement vs Completion

Output: Average engagement is almost the same for both groups (50.30 for non-completers vs 50.67 for completers), so engagement score does not separate students who finish from those who do not.

completion_flag	avg_engagement
0 (Not completed)	50.30
1 (Completed)	50.67

8. SQL File Name:

STAGE 5 -Power BI Dashboard

6.1 KPIs

- Overall Total Students – 18.601K
- Overall Average Engagement Score – 50.50
- Overall Completion Rate – 53.69%
- Average Learning time – 249.34 min per day

6.2 Dashboard Visuals

- Completion vs Dropout Risk
- Student Usage by Device
- Student Distribution by Location
- Engagement Trend Over Time
- Student Distribution by Subscription Type
- Course Completion Status
- Referral Source

6.3 Dashboard Slicers

The dashboard includes interactive slicers for:

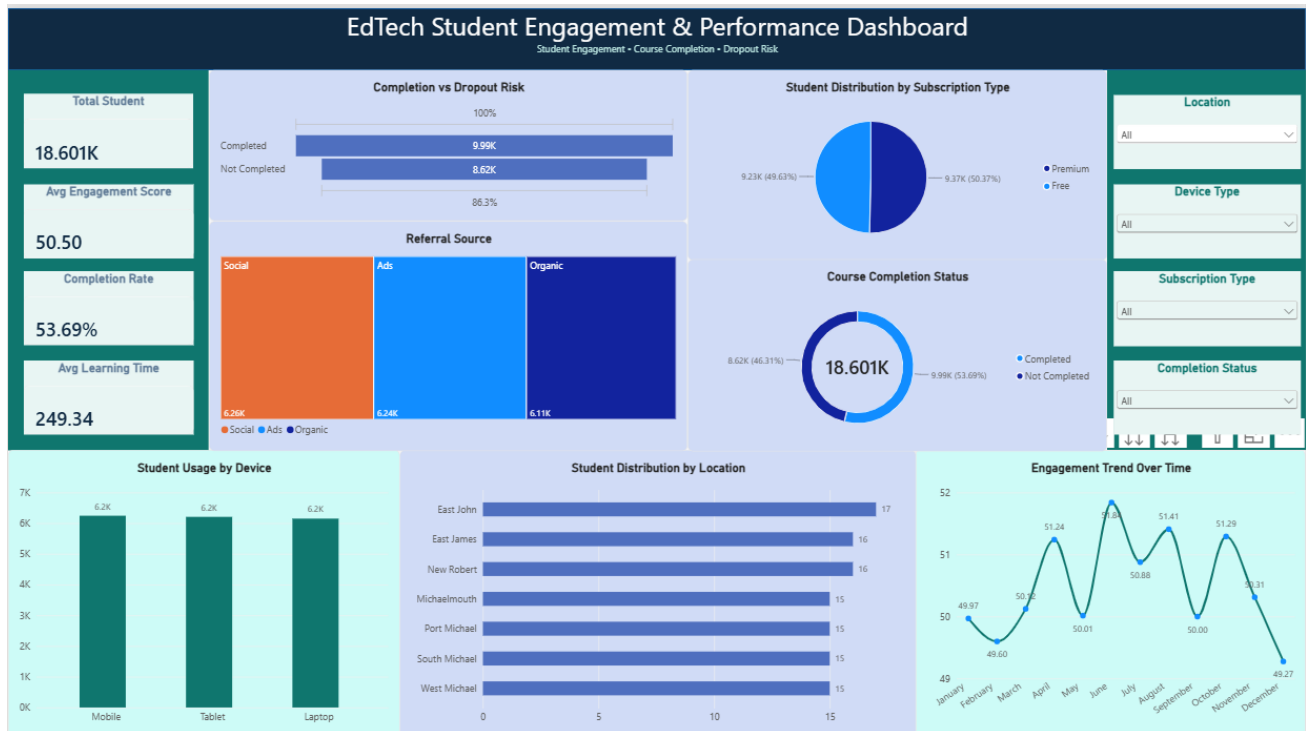
- Location
- Device Type
- Subscription Type
- Completion Status

These slicers allow users to dynamically filter the dashboard and analyse specific student groups based on location, device usage, subscription type, and course completion status.

6.4 Suggested Dashboard Title

EdTech Learning Analytics Dashboard

6.5 Current Status



The Power BI dashboard was created using the cleaned EdTech student activity data loaded into MySQL. It includes KPI cards for total students, average engagement score, completion rate, and dropout risk rate. Interactive visualizations were created to analyse completion and dropout risk, device usage, student location, and engagement trends over time. Slicers for Location, Device Type, Subscription Type, and Completion Status were added to enable users to dynamically filter and explore student engagement and learning behaviour.

6.6 Dashboard File Name

edtech_dashboard.pbix

Stage 7: Insights & Recommendations

7.1 Key Insights

- **Overall Student Engagement:** The EdTech dataset contains **18.601K students**, with an overall **average engagement score of 50.50**, indicating a moderate level of student engagement on the platform.
- **Course Completion:** The overall **completion rate is 53.69%**, showing that slightly more than half of the students completed their courses.
- **Dropout Risk:** Approximately **36.62% of students are classified as being at risk of dropout**, based on the project-defined dropout risk threshold of **61 or above**.

7.2 Recommendations

- **Improve Student Engagement:** The platform can focus on increasing engagement through personalized learning content, reminders, progress notifications, quizzes, and interactive learning activities.
- **Focus on At-Risk Students:** Students with higher dropout-risk scores can be identified early and provided with personalized support, mentoring, reminders, or additional learning resources.
- **Improve Course Completion:** Since the completion rate is 53.69%, the platform can analyse inactive students, assignment submissions, quiz participation, and learning activity to identify barriers to course completion.
- **Optimize Mobile Experience:** Since Mobile is the most frequently used device, the platform should ensure that courses, videos, quizzes, and other learning features provide a smooth mobile experience.

Final Conclusion

The **EdTech Student Engagement Analysis** project transformed raw student data into meaningful insights using Python, MySQL, and Power BI. It helped identify student engagement, course completion, and dropout-risk patterns, providing a clear dashboard for data-driven decisions to improve student learning and retention.